



SwinTExCo: Exemplar-based video colorization using Swin Transformer

Duong Thanh Tran^a, Nguyen Doan Hieu Nguyen^a, Trung Thanh Pham^a, Phuong-Nam Tran^b,
Thuy-Duong Thi Vu^a, Cuong Tuan Nguyen^c, Hanh Dang-Ngoc^d, Duc Ngoc Minh Dang^{a,*}

^a AiTA Lab, Department of Computing Fundamental, FPT University, D1 Street, Saigon Hi-tech Park, Long Thanh My Ward, Thu Duc, 71216, Ho Chi Minh City, Viet Nam

^b Department of Computer Science and Engineering, Kyung Hee University, Yongin, 446-701, Suwon, Republic of Korea

^c Faculty of Engineering, Vietnamese-German University, Ring road 4, Quarter 4, Thoi Hoa Ward, Ben Cat, 75000, Binh Duong, Viet Nam

^d Faculty of Electrical and Electronics Engineering, Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, 268 Ly Thuong Kiet, District 10, 72506, Ho Chi Minh City, Viet Nam

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ABSTRACT

Video colorization represents a compelling domain within the field of Computer Vision. The traditional approach in this field relies on Convolutional Neural Networks (CNNs) to extract features from each video frame and employs a recurrent network to learn information between video frames. While demonstrating considerable success in colorization, most traditional CNNs suffer from a limited receptive field size, capturing local information within a fixed-sized window. Consequently, they struggle to directly grasp long-range dependencies or pixel relationships that span large image or video frame areas. To address this limitation, recent advancements in the field have leveraged Vision Transformer (ViT) and their variants to enhance performance. This article introduces Swin Transformer Exemplar-based Video Colorization (SwinTExCo), an end-to-end model for the video colorization process that incorporates the Swin Transformer architecture as the backbone. The experimental results demonstrate that our proposed method outperforms many other state-of-the-art methods in both quantitative and qualitative metrics. The achievements of this research have significant implications for the domain of documentary and history video restoration, contributing to the broader goal of preserving cultural heritage and facilitating a deeper understanding of historical events through enhanced audiovisual materials.

1. Introduction

Deep-learning approaches for video colorization involve training a model to predict colors for each image frame based on color-semantic relationships from large datasets (Limmer & Lensch, 2016; Vitoria, Raad, & Ballester, 2020; Wu et al., 2021). In these approaches, the primary challenge lies in maintaining temporal consistency among colorized frames; failure to achieve this consistency results in flickering and causes discontinuities in video colorization. The challenge of maintaining temporal consistency has led to the development of exemplar-based video colorization (He, Chen, Liao, Sander, & Yuan, 2018; Li, Lai, John, & Rosin, 2019; Li, Sheng, Li, Ali, & Chen, 2021). In these approaches, a reference image is employed to guide the colorization for each frame. Temporal consistency can be implicitly learned from the color-semantic relationships in the training datasets. More recently, promising results have been achieved by explicitly enforcing temporal

consistency through a recurrent framework, such as propagating colors from the first to the last frame or applying constraints on the temporal consistency between consecutive colorized frames (Zhang et al., 2019). The deep-learning approach often employs CNNs for learning color-semantic relationships from images (LeCun, Bengio et al., 1995). Nevertheless, CNNs are constrained by limited receptive field size, capturing only local information within a fixed window. This limitation makes it challenging for them to grasp large spatial dependencies or pixel relationships across extensive areas of images. In recent years, the Transformer architecture has gained significant popularity among researchers (Vaswani et al., 2017). Transformer models designed for computer vision tasks, such as the ViT (Dosovitskiy et al., 2021) and its variants, have played a pivotal role in overcoming the limitations of CNNs. In this research, we introduce a new architecture applying

* Corresponding author.

E-mail addresses: duongttse160185@fpt.edu.vn (D.T. Tran), nguyenndhse161261@fpt.edu.vn (N.D.H. Nguyen), trungptse160030@fpt.edu.vn (T.T. Pham), tpnam0901@khu.ac.kr (P.-N. Tran), duongvtt9@fe.edu.vn (T.-D.T. Vu), cuong.nt2@vgu.edu.vn (C.T. Nguyen), hanhhdn@hcmut.edu.vn (H. Dang-Ngoc), ducdnm2@fe.edu.vn (D.N.M. Dang).

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Swin Transformer in Exemplar-based Video Colorization (SwinTexCo), an end-to-end model for the video colorization process incorporating the Swin Transformer architecture as the backbone.

1.1. Related works

The field of video colorization encompasses three main approaches: Interactive Colorization, Exemplar-based Colorization, and Fully Automatic Colorization. Interactive Colorization involves user-guided techniques, Exemplar-based Colorization utilizes reference images to propagate colors, and Fully Automatic Colorization aims to automate the process without user intervention. Each approach has its strengths, limitations, and notable research contributions. Understanding these approaches provides valuable insights into the evolution and current state of video colorization techniques.

1.1.1. Interactive colorization

One common approach for colorizing videos involves user-provided tips. Various methods are employed to offer hints to the model, such as using color strokes, scribbles, or points. These hints assist the model in understanding the correspondence between different parts of the image and their respective colors. There are a lot of models using this approach because of its simplicity compared to other approaches. This approach operates on the assumption that pixels belonging to the same object share the same colors, a simple yet effective strategy for providing suitable colors to the image. (Chen, Zou, Zhao, & Tan, 2012) found K-nearest neighbors of each feature pixel to create a segment of colors from the user's input scribbles. (Levin, Lischinski, & Weiss, 2004) and (Yatziv & Sapiro, 2006) used some weighting functions to minimize the difference between the color of adjacent pixels in the YUV color space and YCbCr color space, respectively. SVCNet (Zhao, Po, Liu et al., 2023) used two sub-networks: a color propagation sub-network that performed scribble-based image colorization and a spatiotemporal smoothing sub-network that post-processed the output of the color propagation sub-network. (Jacob & Gupta, 2009) segmented the image using the rainwater simulation technique of watershed segmentation (Osma-Ruiz, Godino-Llorente, Sáenz-Lechón, & Gómez-Vilda, 2007), merging operations through flooding to remove unwanted segments. (Zhang et al., 2017) used a deep neural network, given the grayscale version and user inputs by the two variants: Local Hints Network, which uses sparse user points, and Global Hints Network, which uses global statistics. iColoriT (Yun, Lee, Park, & Choo, 2023) employed ViT (Dosovitskiy et al., 2021) to achieve a global receptive field for propagating user hints across the image. Despite its effectiveness, there has been limited research utilizing this approach for video colorization. This is primarily due to the significant time and human effort required for users to guide the model through each video frame. Moreover, the user-provided hints may only sometimes be accurate, especially considering subtle changes in the video that users might not have been aware of.

1.1.2. Fully automatic colorization

Another approach for colorizing grayscale images involves training a deep neural network, such as multiple layers of CNNs, to store information about the colors of each object in the image (Limmer & Lensch, 2016). Given the characteristics of the research domain, like image generation, image colorization has also been implemented using the Generative Adversarial Networks (GANs) (Vitoria et al., 2020; Wu et al., 2021). With the introduction of ViT (Dosovitskiy et al., 2021) and its variants, several models have utilized this backbone to enhance colorization performance. For instance, in (Wan, Zhang, Chen, & Liao, 2022), the Swin Transformer (Liu et al., 2021) was applied. Initially successful for image colorization, these models have been extended to handle grayscale videos. GANs have been employed for video colorization in (Kouzougliadis, Sfikas, & Nikou, 2019) and (Zhao, Po, Yu et al., 2023). Novel techniques, such as self-regularization (Lei & Chen,

2019), have been introduced in this approach, enabling the models to produce realistic colors. However, to achieve optimal results, these models require a deep architecture and an extensive training dataset to learn the intricate mapping of colors to different parts of the video frames. Moreover, this approach has limitations on the customization ability of the colorizing process.

1.1.3. Exemplar-based colorization

Another method for colorizing images and videos involves transferring colors from a reference image or a reference video sequence to a grayscale or video frame. The reference image selection is often based on cosine similarity or Euclidean distance. The initial deep-learning model for this approach employed CNNs (LeCun et al., 1995), explicitly using the VGG19 backbone (Simonyan & Zisserman, 2015) to extract colors from the grayscale image and the reference image. This method saw rapid development with the introduction of several enhanced techniques, including the cross-scale matching technique (Li et al., 2019) and the integration of GANs (Li et al., 2021). A model known as Deep exemplar-based video colorization was introduced as an architecture for colorizing videos based on an example image (Zhang et al., 2019). This model incorporated a recurrent architecture to facilitate information sharing between video frames. However, this architecture allowed information to flow in only one direction, from the first frame to the last, potentially impacting the model's performance. To address this limitation, the authors in BiSTNet (Yang et al., 2024) introduced bidirectional recurrent work to enhance information exchange between frames. Another approach dealing with that problem was Chen et al.'s approach (Chen et al., 2024), which used a double-head non-local operation followed by a parallelized CNN-Transformer block to improve the long-term performance of the colorization process. This method is a part of the TCVC model (Zhang et al., 2023), with the addition of a Reference Colorization Network utilized to colorize the first video frame using the fully automatic approach, then Chen et al.'s approach (Chen et al., 2024) acted as the "stage 2" of the pipeline to colorize the remaining video frames. Addressing local issues encountered in CNNs by harnessing the Vision Transformer (ViT) (Dosovitskiy et al., 2021) architecture, Vitexco (Tran et al., 2023) adeptly captures global features from the video frame, thereby enhancing the precision of the colorization process. Besides transferring the colors from a reference image, other approaches, such as Xia et al.'s approach (Xia, Liu, Fang, Yang, & Guo, 2016), used a video sequence as the input for the model. Especially, DeepRemaster (Iizuka & Simo-Serra, 2019) leveraged the mechanism of Swin Transformer (Liu et al., 2022) in processing the extracted features from the previous convolutional modules, which may not take advantage of Swin Transformer in extracting global and local features from an image or a video frame. This approach in previous research has shown promising results but has yet to fully address the challenge of blending colors seamlessly, resulting in residual blending issues.

1.2. Motivation

The field of video colorization, while having seen significant advancements, still faces notable challenges. Each of the three main approaches mentioned above has limitations restricting their effectiveness. For instance, Interactive Colorization is time-consuming due to the significant amount of human input required. Despite recent techniques such as cross-scale matching (Li et al., 2019) and GAN integrations (Li et al., 2021), exemplar-based colorization needs to improve on blending issues. Fully Automatic colorization, while promising, requires extensive training datasets and very deep architectures to be fully effective (Limmer & Lensch, 2016; Wu et al., 2021).

Recent advancements in deep learning techniques provide fresh avenues for addressing these persistent challenges. The Vision Transformer (ViT) architecture (Dosovitskiy et al., 2021) and its variants, for instance, have been found to exceptionally capture global features

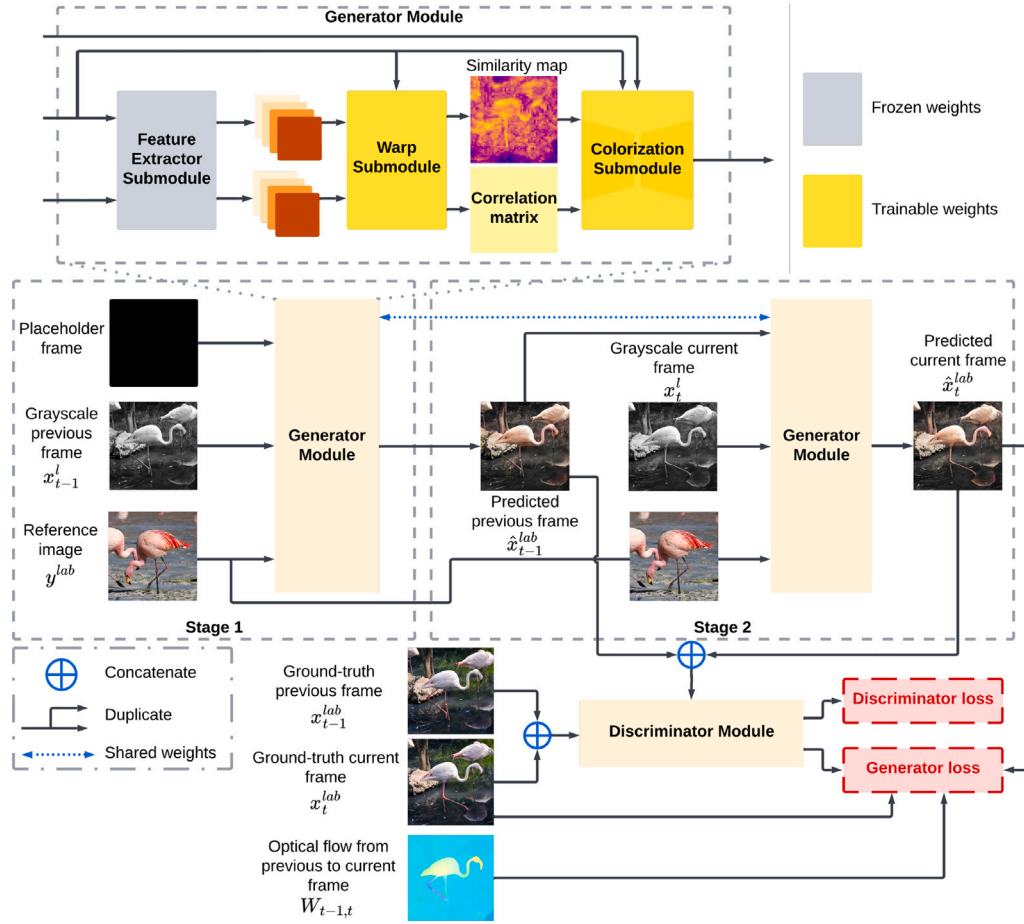


Fig. 1. The overall architecture of SwinTexCo.

in an image or video frame (Tran et al., 2023). Other innovations such as GANs (Vitoria et al., 2020; Zhao, Po, Yu et al., 2023) and techniques like self-regularization (Lei & Chen, 2019) have demonstrated potential in generating realistic colors in video colorization.

Our research is motivated by the compelling advantages offered by the Swin Transformer (Liu et al., 2021), a variant of the ViT, to overcome the limitations of previous works. Specifically designed for image and video processing, the Swin Transformer presents a promising solution to many challenges in the field. The model's overlapping window-based self-attention mechanism allows hierarchical representation and processing of image and video data at different scales, thus addressing blending issues typical in exemplar-based colorization. Regarding operational benefits, the Swin Transformer's shifted windows significantly reduce the computational cost of videos, enhancing overall efficiency. Further, the inherent deep architecture of Swin Transformer can eliminate the need for extensive customizations in fully automatic colorization, streamlining the colorization process while improving accuracy.

1.3. Our approach and contributions

To address the above-mentioned challenges in Section 1.1, SwinTexCo utilizes the Swin Transformer (Liu et al., 2021) as an image encoder backbone for the exemplar-based video colorization task. Swin Transformer is a variant of the Vision Transformer, offering advantages through a hierarchical architecture that captures both local and global dependencies. These benefits include improving computational efficiency with shifted windows and providing a more interpretable structure for understanding learned representations. SwinTexCo also integrates the recurrent framework proposed by (Tran et al., 2023) and

(Zhang et al., 2019) for video colorization, enforcing temporal consistency constraints and adversarial learning. The main contributions of our work are summarized below:

- We introduce a novel network architecture utilizing the Swin Transformer backbone to extract local and global features for colorizing videos using an exemplar-based colorization approach. Furthermore, it mitigates the reliance on user-provided hints seen in interactive colorization methodologies while avoiding using extensive models and datasets as observed in fully automatic colorization techniques.
- We tackle the problem of high computational costs that have been encountered in previous models while maintaining the effectiveness of our video colorization approach.
- We generate a large, high-quality dataset specifically for training the model. This dataset comprises a diverse range of categories amalgamated from various standard datasets.
- We evaluate our network architecture using quantitative and qualitative evaluation metrics to demonstrate the competitive outcomes achieved compared to numerous superior models.

Our research adopts a structured approach to effectively present the research on video colorization. Section 2 presents our proposed methodology for this research, detailing the techniques, algorithms, and models used in the colorization process. Section 3 describes the training process and the loss functions used in the training phase. The preliminary results obtained from experiments are then presented and analyzed in Sections 4 and 5, using various metrics and criteria to assess the quality and accuracy of the colorization outcomes. The research culminates in Section 6, which summarizes the key findings, discusses

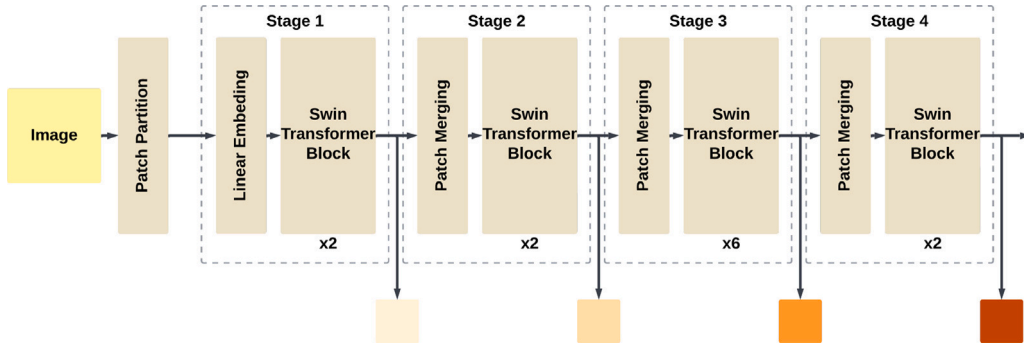


Fig. 2. The architecture of SwinT Extractor.

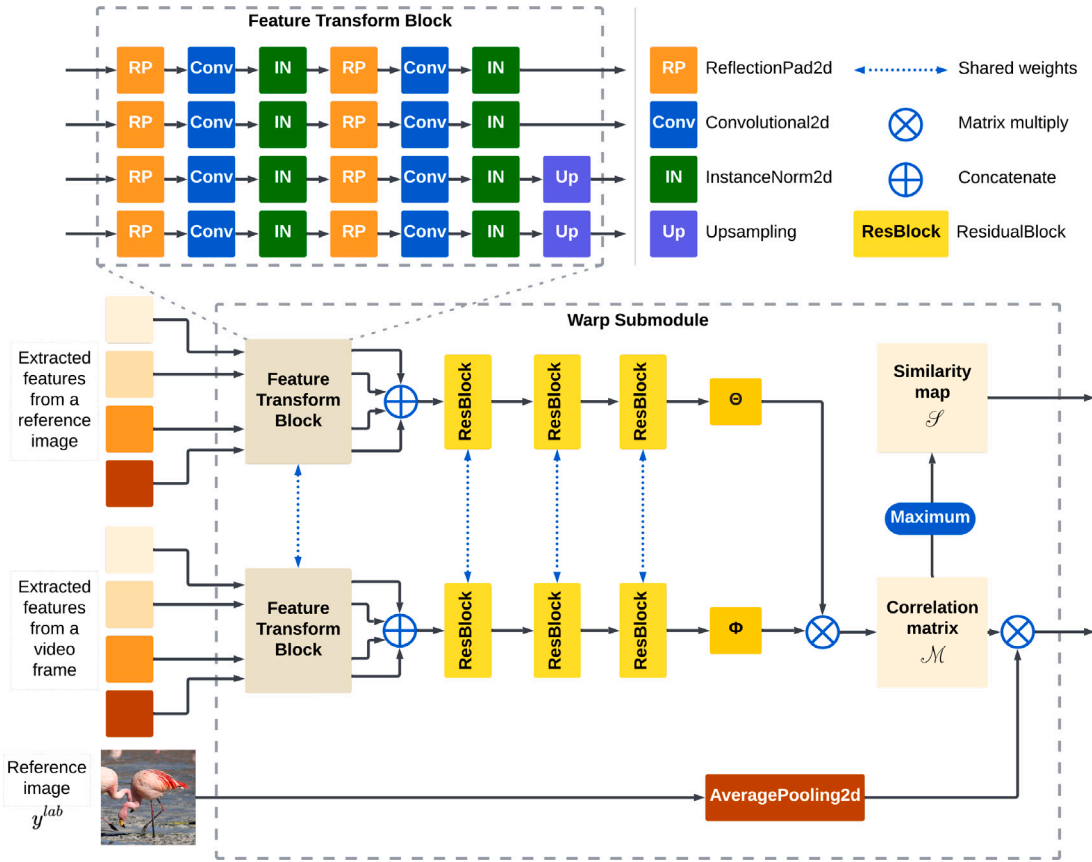


Fig. 3. The architecture of Warp submodule.

the implications, and highlights potential future research avenues in video colorization.

2. The proposed method

The overall network architecture of SwinTexCo is illustrated in Fig. 1. The model comprises two main components: the Generator Module and the Discriminator Module. The Generator Module comprises three distinct submodules, each serving a specific purpose. First, the SwinT Extractor Submodule extracts feature from the video frame and the reference image, leveraging the Swin Transformer architecture to effectively capture relevant visual information. Second, the Warp Submodule generates a Correlation Matrix and a Similarity Map based on the extracted features from the SwinT Extractor Submodule. This component is crucial for establishing meaningful relationships between the video frame and the reference image. Finally, at the core of the

colorization process, the Colorization Submodule utilizes the Correlation Matrix and Similarity Map to effectively add color to the video frames. On the other hand, the Discriminator Module plays a pivotal role in evaluating the quality of the colorization process performed by the Generator Module. Employing an adversarial training scheme, the Discriminator Module provides feedback to guide and refine the colorization process, ensuring the generation of visually pleasing and realistic colored videos.

2.1. Overall network

Our approach employs the LAB color space, which comprises three channels: l , a , and b . The channel l represents perceptual lightness, with values within the $[-100, 100]$ range. The remaining two channels, a and b , correspond to the red/green and blue/yellow values within the $[-128, 128]$ range. Leveraging the LAB color space, SwinTexCo's colorization process only needs to adjust the a and b channels while

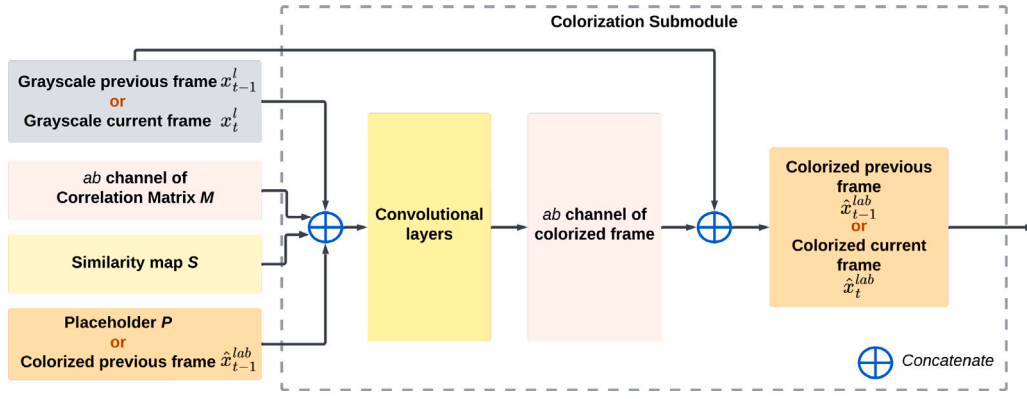


Fig. 4. The architecture of Colorization submodule.

preserving the contextual information of the l -channel in video frames. Fig. 1 provides an overview of the architectural design. Let $X^l = \{x_t^l\}, t = 1, T$ denotes an input video in the l -channel (grayscale video) consisting of T frames, with $x_t^l \in \mathbb{R}^{H \times W \times 1}$ representing the t th video frame. The dimensions $H \times W$ indicate the size of each video frame, which represents the resolution of the video. Our goal is to generate a colorized video $\hat{X}^{lab} = \{\hat{x}_t^{lab}\}, t = 1, T$, where $\hat{x}_t^{lab} \in \mathbb{R}^{H \times W \times 3}$ denotes the colorized frame at time t in the LAB color space. The colorization process utilizes a reference color image $y^{lab} \in \mathbb{R}^{H \times W \times 3}$ for each video and the previous colorized frame x_{t-1}^{lab} as guidance.

2.2. Generator module

The Generator module comprises three submodules: SwinT Extractor, Warp and Colorization. The details of each submodule will be provided in the following sections.

2.2.1. SwinT extractor submodule

The SwinT Extractor submodule employs a pre-trained Swin Transformer (Liu et al., 2021) to extract features from an input image. The extracted features consist of outputs from the four stages of the Swin Transformer. Initially trained on the ImageNet-21K dataset (Ridnik, Ben-Baruch, Noy, & Zelnik-Manor, 2021) and subsequently fine-tuned on the ImageNet-1K dataset (Deng et al., 2009), the Swin Transformer's 4-stage selection aligns with its internal architecture. This process allows us to capture information from the image at different hierarchical levels. In summary, as illustrated in Fig. 2, SwinT Extractor receives a video frame or a reference image as input and performs feature extraction. It generates four tensors representing the extracted features corresponding to the input video frame or image.

2.2.2. Warp submodule

The Warp submodule, depicted in Fig. 3, encompasses a CNN model responsible for generating the Correlation Matrix $\mathcal{M} \in \mathbb{R}^{HW \times HW}$ and the Similarity Map $S \in \mathbb{R}^{H \times W}$ between the grayscale video frame and the reference image. Using the feature tensors extracted from SwinT Extractor, a sequence of layers – such as Convolutional layers, Normalization layers, Padding layers, Upsampling layers, Dropout layers, and activation functions – is employed to transform these feature tensors. The feature tensors initially associated with specific objects are concatenated to produce two tensors: one corresponding to the grayscale video frame, denoted as Θ , and the other representing the reference image, denoted as Φ . The Correlation Matrix between these two tensors is computed through a matrix multiplication operation. Subsequently, the Similarity Map is derived from the Correlation Matrix by applying the Maximum Operator. This Similarity Map also serves as input for the subsequent submodules. To summarize, leveraging the feature tensors extracted from the SwinT Extractor as inputs, the Warp submodule produces a Correlation Matrix $\mathcal{M}$ and a Similarity Map S .

2.2.3. Colorization submodule

The Colorization submodule, displayed in Fig. 4, serves as the central component of the overall model and is tasked with executing the colorization process. It incorporates multiple layers, including Convolutional layers, Normalization layers, and activation functions. In our methodology, this submodule is employed to calculate the discrepancy between the colorized results of the current and previous frames. The aim is to adjust the colors in the predicted current frame, ensuring consistency in colorization. To achieve accurate colorization of the current video frame, the involvement of the previous frame is necessary to maintain color consistency across adjacent frames within the video. In the colorization process within this submodule, two distinct scenarios are encountered. In the first scenario, when the currently colorized frame is the initial frame of the video and therefore lacks a preceding frame, the model employs a Placeholder image denoted as $\mathcal{P}$. This Placeholder image is essentially black. The rationale behind this selection serves twofold from a primary standpoint. Firstly, it offers a simplified approach, necessitating minimal additional programming or processing steps, thereby proving advantageous when dealing with extensive datasets or striving for enhanced speed and efficiency. Secondly, utilizing a black image ensures the model's equilibrium during the normalization of all video frames. In most cases, adopting a black image as the preceding frame represents a sensible and pragmatic approach. The second scenario pertains to situations where the currently colorized video frame is beyond the initial frame. In such cases, the model simply employs the immediately preceding frame of the current frame for the subsequent colorization process. In brief, the Colorization submodule takes the grayscale video frame x_t^l , the Correlation Matrix $\mathcal{M}$, the Similarity Map S , and either the colorized previous frame $\hat{x}_{t-1}^{lab}$ or the black image as inputs. It then generates the colorized current video frame $\hat{x}_t^{lab}$.

2.3. Discriminator module

The quality assessment of the colorization process is performed by utilizing the Discriminator module, shown in Fig. 1. This module is a model that embeds the input tensor into a vector to facilitate subsequent computations. The input tensors consist of the concatenation of the predicted previous video frame and the predicted current video frame, as well as the concatenation of the ground truth previous video frame and the ground truth current video frame. These tensors are passed through the submodule, and multiple loss functions, which will be discussed in Section 3.2, are employed to quantify the similarity between the predicted and ground truth video frames. In essence, the Discriminator plays a pivotal role in discerning the realism of the colorized video frames.

3. Training frameworks

The training process will be initiated based on the modules above and submodules. This section elucidates how the data traverses the comprehensive model and the loss functions imposed on the training process.

3.1. Training process

In each training iteration, as shown in Fig. 1, the colorization task is divided into two stages. These stages are responsible for colorizing two adjacent video frames, ensuring coherent and continuous colors to prevent flickering artifacts. The two stages differ depending on whether it is the first or the latter in a pair of adjacent frames. For convenience of description, the first frame will be called the previous frame, and the latter will be called the current frame. In the first stage, a black image called Placeholder $\mathcal{P}$ is combined with the grayscale previous frame x_{t-1}^l and the reference image y to generate the colorized previous frame $\hat{x}_{t-1}^{lab}$. In the second stage, the output from the first stage $\hat{x}_{t-1}^{lab}$ is combined with the grayscale current frame x_t^l and the reference image y to produce the colorized current frame $\hat{x}_t^{lab}$. The Discriminator module then evaluates these outputs. This process is repeated in subsequent iterations, facilitating the training progress.

3.2. Losses

As depicted in Fig. 1, the training process of our model incorporates two distinct losses, namely the Generator Loss and the Discriminator Loss. The Generator Loss is employed during the backpropagation of the Generator Module (Section 2.2), facilitating the optimization of its parameters. On the other hand, the Discriminator Loss is utilized during the backpropagation of the Discriminator Module (Section 2.3), allowing for the refinement of its distinguishing capabilities. Notably, when backpropagating the Discriminator, the Generator's parameters are frozen to ensure a focused learning process. The subsequent sections will delve deeper into the intricate workings of these two losses, shedding light on their specific roles in enhancing the overall performance of our model.

3.2.1. Generator loss

This research works on an extensive model, which contains a lot of large submodules. Besides, we want to ensure a lot of constraints, for instance, the similarity between a predicted current video frame and its reference image, the consistency between a predicted current video frame and its predicted previous video frame, the smoothness in colorizing a video frame, etc. Therefore, we implement weighted sum multiple Generator Loss functions to constrain them.

L1 loss: L1 Loss function calculates the differences in color between a colorized video frame $\hat{x}_t$ and the corresponding ground truth frame x_t . By minimizing the loss function, the video prediction model can generate more accurate predictions that closely resemble the original video frames. The formula for L_{L1} can be expressed as follows:

$$L_{L1} = \|\hat{x}_t^{lab} - x_t^{lab}\|_1. \quad (1)$$

Adversarial loss: An Adversarial Loss is incorporated as a constraint in the video colorization process to enhance the realism of the colorized videos. In line with the approach adopted in Deep Exemplar-based Video Colorization (Zhang et al., 2019), a video discriminator is employed instead of an image discriminator. This selection is motivated by the fact that the flickering in colorized videos can be detected compared to real videos. Using a video discriminator, the model can learn to colorize the videos while minimizing the flickering issue and ensuring the realism of the colorized video frames. This approach effectively

improves the quality and visual fidelity of the colorized videos. The adversarial loss L_{adv} can be formulated as:

$$L_{adv} = \mathbb{E}_{(\hat{x}_{t-1}, \hat{x}_t)} \left[\left(D(\hat{x}_{t-1}, \hat{x}_t) - \mathbb{E}_{(x_{t-1}, x_t)} D(x_{t-1}, x_t) - 1 \right)^2 \right] + \mathbb{E}_{(x_{t-1}, x_t)} \left[\left(D(x_{t-1}, x_t) - \mathbb{E}_{(\hat{x}_{t-1}, \hat{x}_t)} D(\hat{x}_{t-1}, \hat{x}_t) + 1 \right)^2 \right]. \quad (2)$$

$D(x_{t-1}, x_t)$ denotes passing the previous video frame and current video frame to the Discriminator Module (Section 2.3). The result from $D(x_{t-1}, x_t)$ is a scalar floating point value ranging within [0,1]. The scalar value represents the probability that the input frames are real frames. The expression $\mathbb{E}_{(x_1, x_2)} F(x_1, x_2)$ represents the expectation of the function $F(x_1, x_2)$ with respect to the joint probability distribution of the variables (x_1, x_2) .

Perceptual loss: The Perceptual Loss, which was first introduced in (Gatys, Ecker, & Bethge, 2015), is used for training a neural network to transfer the style of an image to another in a semantic manner. It measures the difference between the high-level features of 2 images, such as edges, textures, and shapes. We use this loss to make the output perceptually plausible. This loss penalizes the semantic difference between the predicted video frame $\hat{x}_t$ and the correspondent ground truth frame x_t . The formula for L_{perc} can be expressed as follow:

$$L_{perc} = \|\phi_{\hat{x}}^s - \phi_x^s\|_2^2, \quad (3)$$

where $\phi_{\hat{x}}^s$ represents the output feature at s th stage of SwinT Extractor with the input $\hat{x}$.

Contextual loss: Contextual Loss measures the difference between the high-level features of 2 images, considering the context of the entire image and edges, textures, and shapes. In (Mechrez, Talmi, & Zelnik-Manor, 2018), the contextual loss was used to train a neural network to transfer the style of an image to another while preserving the overall structure and meaning. This allows the neural network to learn how to transform an input image into an output image with the desired features. Given its ability to compare contextual meaning between a video frame and a reference image, the contextual loss is suitable for encouraging the colors in the predicted frame $\hat{x}_t$ to be similar to those in the corresponding reference image. We first compute the cosine similarity $d^s(i, j)$ between each pair of feature points $\phi_{\hat{x}}^s(i)$ and $\phi_y^s(j)$ and $\hat{d}^s(i, j) = d^s(i, j) / (\min_k d^s(i, k) + \epsilon)$ where $\epsilon = 10^{-5}$. We have the formula for pairwise affinities $A^s(i, j)$ across features:

$$A^s(i, j) = \text{softmax}_j \left(1 - \hat{d}^s(i, j)/h \right), \quad (4)$$

where $h = 0.1$ is the bandwidth parameter (Zhang et al., 2019), The value of $A^s(i, j)$ distribute in range [0,1] show the similarity between $\hat{x}_t(i)$ and $y(j)$ in the s th stage features. On the basis of forward matching in (Zhang et al., 2019), for each feature $\phi_{\hat{x}}^s(i)$ we find the most relevant feature $\phi_y^s(j)$ in y . In summary, the purpose of contextual loss is to maximize the contextual similarity between the predicted frame and the reference by $A^s(i, j)$:

$$L_{ctx} = \sum_s w_s \left[-\log \left(\frac{1}{N_s} \sum_i \max_j A^s(i, j) \right) \right]. \quad (5)$$

We use 4 feature maps: $s = 1$ to 4, in which N_s indicates the feature number of s th stage from SwinT Extractor, and w_s is the suitable weight for s th stage.

Temporal consistency loss: A temporal consistency loss (Chen, Liao et al., 2017) is incorporated into the colorization process to account for temporal coherence in video colorization. This loss function explicitly penalizes color changes that occur along the flow trajectory. By incorporating this constraint, the model can ensure that the color changes in the video frames are consistent over time, thereby improving the overall visual quality and coherency of the colorized video. This approach is particularly effective in addressing flickering in video colorization tasks, as it encourages the model to produce smoother and more

natural-looking color transitions. To reinforce temporal consistency in the model, we utilize the RAFT algorithm (Teed & Deng, 2020) to generate the optical flow information of each pair and method (Ruder, Dosovitskiy, & Brox, 2016) to generate an occlusion mask. By leveraging these flow and mask components, we calculate the loss and improve the overall visual quality of the output. The formula for L_{temp} can be expressed as follow:

$$L_{temp} = \|m_{t-1} \odot W_{t-1,t}(\hat{x}_{t-1}^{lab}) - m_{t-1} \odot \hat{x}_t^{lab}\|, \quad (6)$$

where $W_{t-1,t}$ denotes the forward flow from the previous frame x_{t-1} to x_t and m_{t-1} is the binary mask and $\odot$ denotes Hadamard product operation.

Smoothness loss. Smoothness Loss was also introduced in (Zhang et al., 2019) to encourage spatial smoothness. In video colorization tasks, it is often assumed that neighboring pixels of the predicted frame $\hat{x}_t$ should have similar chrominance values if they have similar chrominance in the corresponding ground truth frame x_t . A smoothness loss is incorporated into the colorization process to enforce this constraint. This loss function calculates the difference between the current pixel's color and the weighted color of its 8-connected neighborhood. By minimizing this loss, the model can ensure that the color transitions in the video frames are smooth and consistent, thus improving the overall visual quality of the colorized video. This approach is particularly effective in producing natural-looking color transitions and reducing the occurrence of color artifacts in the colorized video. The formula for L_{smooth} can be expressed as follows:

$$L_{smooth} = \frac{1}{N} \sum_{c \in \{a,b\}} \sum_i \left(\hat{x}_t^c(i) - \sum_{j \in \mathcal{N}(i)} w_{i,j} \hat{x}_t^c(j) \right), \quad (7)$$

where N is the number of samples in a training step. $w_{i,j}$ is weighted least squares (WLS) weight indicating correlations between nearby pixels.

Weighted-sum generator loss. We add all the losses above to create the total loss that we want to optimize:

$$L = \lambda_{L1} L_{L1} + \lambda_{adv} L_{adv} + \lambda_{perc} L_{perc} + \lambda_{ctx} L_{ctx} + \lambda_{temp} L_{temp} + \lambda_{smooth} L_{smooth}, \quad (8)$$

where λ is the weight of each loss function contribution to the total loss formula.

3.2.2. Discriminator loss

The Discriminator Loss is used for the backpropagation process of the Discriminator Module. Its primary purpose is to enhance the Discriminator's ability to differentiate between a ground truth frame and a colorized frame. The Discriminator Loss L^D can be calculated as follow:

$$L^D = \frac{1}{2} \left(\mathbb{E}_{(\hat{x}_{t-1}, \hat{x}_t)} \left[\left(D(\hat{x}_{t-1}, \hat{x}_t) - \mathbb{E}_{(x_{t-1}, x_t)} D(x_{t-1}, x_t) + 1 \right)^2 \right] + \mathbb{E}_{(x_{t-1}, x_t)} \left[\left(D(x_{t-1}, x_t) - \mathbb{E}_{(\hat{x}_{t-1}, \hat{x}_t)} D(\hat{x}_{t-1}, \hat{x}_t) - 1 \right)^2 \right] \right). \quad (9)$$

The Discriminator Loss exhibits a comparable characteristic to the Adversarial Loss (Section 3.2.1) implemented for the Generator (Section 2.2) but differs slightly in its operator. This discrepancy in the operator choice leads to the gradient tensor being in the opposite direction compared to the Generator's loss. By optimizing the Discriminator Loss, we aim to train the Discriminator (Section 2.3) to become more proficient in discerning between ground truth frames and colorized frames, ultimately improving the overall performance of the colorization process.

Table 1

Summary of training datasets.

Dataset name	#videos	#frames
Hollywood2 (Marszalek et al., 2009)	269	12,560
Pixabay (Pixabay, 2023)	783	38,416
DAVIS (Pont-Tuset et al., 2017)	90	6,118
REDS (Nah et al., 2019)	300	23,760
NTIRE 2023 (Kang et al., 2023)	170	21,178
Augmented dataset from ImageNet-1K (Deng et al., 2009)	–	70,000
Total	1612	172,032

4. Implementation details

4.1. Dataset

4.1.1. Training dataset

We collect a comprehensive collection of video datasets specifically tailored for training. Our dataset encompasses a diverse range of sources, including the Hollywood2 (Marszalek, Laptev, & Schmid, 2009), DAVIS (Pont-Tuset et al., 2017), REDS (Nah et al., 2019), videos sourced from Pixabay (Pixabay, 2023), and the training dataset from the NTIRE 2023 Video Colorization Challenge (Kang et al., 2023). The combined dataset comprises 269 videos from the Hollywood2 dataset, 783 videos downloaded from Pixabay using the Pixabay API, and contributions from the DAVIS dataset, which offers densely annotated video object segmentation data. Additionally, we leverage the REDS dataset, which serves as a valuable resource for video deblurring and super-resolution tasks. Furthermore, we incorporate the NTIRE 2023 Video Colorization Dataset, which consists of video sequences specifically created for the video colorization domain.

By applying cosine similarity search on feature embeddings of images, we curate the k_{ref} most similar images from the ImageNet-1K dataset (Deng et al., 2009) to each pair of the current and previous frame (detailed in Table 2). These images serve as a reference to guide the model in colorization. To enhance the model's ability to transfer color patterns from reference images to the colorized output, each time a sample is used for training, the reference image could either be the ground truth previous frame, the ground truth current frame, or a randomly chosen image from the set of k_{ref} images mentioned earlier. The probability of using one of k_{ref} images as a reference image denotes P_{ref} . This research uses the RAFT algorithm (Teed & Deng, 2020) for extracting optical flow information. Fig. 5 shows a sample of the training dataset.

To enhance the diversity of the dataset, we randomly selected 70,000 images from the ImageNet-1K dataset (Deng et al., 2009), ensuring that each image had a corresponding reference image. We apply random geometric distortion and luminance noise techniques to introduce greater variety and realism into the dataset. This augmentation process can be observed in Fig. 6, where the resulting video frames display enhanced diversity and increased realism. Augmented *Current frame* (middle) created by applying random geometric distortion and luminance noise to the image. *Previous frame* (left) represents the original unaltered image. *Reference image* (right) also obtained from ImageNet-1K.

We ensure the quality of our dataset through a filtering process, eliminating low-quality and black-and-white videos from the sources above. Additionally, we conduct a rigorous manual verification process to ensure adherence to our quality standards. Consequently, we accumulated 172,032 samples for the training dataset, summarized in Table 1.

4.1.2. Evaluation dataset

We utilize the DAVIS dataset (Perazzi et al., 2016), a widely-used benchmark dataset in computer vision. It encompasses a diverse collection of 30 high-resolution videos, enabling researchers to evaluate

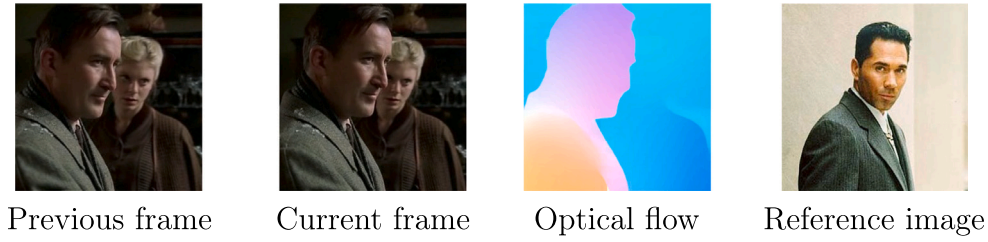


Fig. 5. Example from the training dataset.



Fig. 6. Example from augmented ImageNet-1K.

Table 2

Hyper-parameters settings in training model.

Hyper-parameter	Description	Value
γ_0	Learning rate	3×10^{-5}
β_1	First exponential decay rate	0.500
β_2	Second exponential decay rate	0.999
λ_{L1}	L1 loss's weight	2.000
λ_{adv}	Adversarial loss's weight	0.500
λ_{perc}	Perceptual loss's weight	0.100
λ_{ctx}	Contextual loss's weight	1.000
λ_{temp}	Temporal loss's weight	0.300
λ_{smooth}	Smoothness loss's weight	5.000
k_{ref}	Number of candidate reference images	5
P_{ref}	Probability of using reference image from ImageNet-1k	0.95

and compare the performance of different video colorization models. Additionally, we incorporate 20 videos from the Videvo dataset (Lai et al., 2018; Stival, 2023). Each video comprises over 200 frames, presenting a challenging scenario for model evaluation.

4.2. Hyper-parameters

We utilize a modified version of SwinV2-T (Liu et al., 2022), the tiny pre-trained version of the SwinV2 model, as the backbone of SwinTexCo architecture. The weights assigned to the losses in the model are set as in Table 2. For optimization, we employed the AdamW optimizer (Loshchilov & Hutter, 2019) with an initial learning rate γ_0 , and exponential decay rate β_1 and β_2 of first and second-moment estimates, for both the Generator Module and Discriminator Module. We also adopt the “poly” learning rate decay inspired by the work of DeepLab (Chen, Papandreou et al., 2017):

$$\gamma = \gamma_0 \left(1 - \frac{N_{iter}}{N_{total}}\right)^{0.9}, \quad (10)$$

where N_{iter} and N_{total} represent respectively the current and total iteration numbers. The model is trained for 20 epochs on 2 NVIDIA RTX A6000 GPUs using a batch size of 4. We also implement the distributed data-parallel technique to increase the training speed.

4.3. Inference stage

In the inference stage, different from all training steps mentioned in Section 3, the process involves utilizing the model to colorize the entire

video. This stage comprises several distinct steps aimed at achieving accurate and efficient colorization. Initially, the video is divided into separate scenes using the PySceneDetect tool (Castellano, 2018). Next, a reference image is identified for each scene to facilitate the colorization process. Subsequently, the colorization procedure is applied to all scenes using the corresponding reference images.

To enhance the efficiency of the reference image search, we incorporate a high-performance vector database. This database leverages the Faiss (Johnson, Douze, & Jégou, 2021) library for expedited searching based on cosine similarity. By utilizing the advanced searching mechanism, we can rapidly retrieve the most relevant reference images for colorization.

5. Experimental results

5.1. Evaluation metrics

5.1.1. Quantitative metrics

The evaluation of SwinTexCo's performance can be difficult, given the subjective nature of aesthetics in colorized images and videos. Nevertheless, several assessments were used to evaluate the model. The following standard metrics were used to evaluate the colorized video:

- Peak signal-to-noise ratio (PSNR): This metric measures the difference between the original signal and a compressed or transmitted version of that signal, considering the amount of noise introduced during compression or transmission. In this research, PSNR was used to measure the amount of signal (i.e., colors) in the colorized video frames.
- Structural Similarity Index Measure (SSIM) (Wang, Bovik, Sheikh, & Simoncelli, 2004): SSIM measures the similarity between 2 images. It is based on the principle that the quality of an image as perceived by humans is closely related to the structural information it contains. It considers the luminance, contrast, and structural similarity between a reference image and a distorted image. This research used SSIM to compare each colorized video frame to its ground truth.
- Learned Perceptual Image Patch Similarity (LPIPS) (Zhang, Isola, Efros, Shechtman, & Wang, 2018): Unlike traditional metrics such as PSNR and SSIM, which rely on crafted features and assumptions about the human visual system, LPIPS is based on a deep neural network that was trained on a large dataset of images to learn features relevant to human perception. LPIPS was used to measure the similarity between each colorized video frame and its ground truth.

Table 3

Comparison with state-of-the-art video colorization models on DAVIS (Pont-Tuset et al., 2017) and Videvo (Lai et al., 2018) datasets about quantitative metrics.

Models	DAVIS dataset				
	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	CDC ↓
FAVC (Lei & Chen, 2019)	16.80765	0.49951	0.36293	181.54643	0.00339
Color2Embed (Zhao, Wu, Liu, & He, 2021)	27.09984	0.92238	0.12797	97.70031	0.00381
VCGAN (Zhao, Po, Yu et al., 2023)	23.74646	0.89443	0.17274	146.69747	0.00828
DeepExemplar (Zhang et al., 2019)	29.63041	0.94850	0.08847	73.30805	0.00451
DeepRemaster (Iizuka & Simo-Serra, 2019)	24.90564	0.87459	0.20152	148.78267	0.00934
TCVC (Zhang et al., 2023)	31.81122	0.96309	0.06695	58.48240	0.00392
Vitexo (Tran et al., 2023)	24.53519	0.93877	0.13591	150.32256	0.00433
SwinTexCo (ours)	32.24419	0.96586	0.06328	56.66980	0.00398
Models	Videvo dataset				
	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	CDC ↓
FAVC (Lei & Chen, 2019)	17.47619	0.54666	0.36249	153.20350	0.00163
Color2Embed (Zhao et al., 2021)	27.25240	0.93831	0.14053	97.58350	0.00177
VCGAN (Zhao, Po, Yu et al., 2023)	25.02977	0.92525	0.18520	119.60610	0.00300
DeepExemplar (Zhang et al., 2019)	29.75574	0.96154	0.10054	77.69207	0.00192
DeepRemaster (Iizuka & Simo-Serra, 2019)	25.38152	0.90306	0.18771	119.39434	0.00290
TCVC (Zhang et al., 2023)	31.19570	0.97091	0.07714	65.11399	0.00168
Vitexo (Tran et al., 2023)	25.36680	0.95398	0.13882	136.06062	0.00200
SwinTexCo (ours)	31.96999	0.94401	0.10871	43.36435	0.00172

- Fréchet inception distance (FID) (Heusel, Ramsauer, Unterthiner, Nessler, & Hochreiter, 2017): This metric is generally used to assess the quality and diversity of images generated by GANs. It uses a pre-trained Inception network to extract feature vectors from ground truth and colorized frames and then calculates the mean and covariance of these vectors. The FID is computed as the Fréchet distance between the distributions these statistics define. Instead of measuring the distance between a ground truth and a colorized frame, we used FID to measure the distance between each colorized video frame and its ground truth.
- Color Distribution Consistency (CDC) (Liu et al., 2024): CDC is commonly used to assess the similarity of color distributions between 2 images or video frames. It considers the distribution of colors within an image or frame rather than focusing solely on individual pixel values or overall image quality. It evaluates the consistency of color distribution by comparing the frequencies of different color values across the image or frame.

The metrics above are computed frame by frame for the predicted video, with their mean value serving as the overall evaluation. We continue this process for the entire evaluation dataset and calculate the mean to arrive at the final value.

5.1.2. Qualitative metrics

Although quantitative metrics are limited in their ability to capture human sensory perception fully, we conducted a survey utilizing the Mean Opinion Score (MOS) metric to evaluate human perception. For comparison, we select two models, namely DeepExemplar (Zhang et al., 2019) and TCVC (Zhang et al., 2023), because they also employed an exemplar-based approach for video colorization, which aligns with our proposal to ensure a fair and relevant comparison.

5.2. Results and discussions

5.2.1. Quantitative results

We conduct quantitative evaluations thoroughly described in Section 5.1. As presented in Table 3, the results of our assessments provide valuable insights into each model. Table 3 shows that SwinTexCo demonstrates superior performance to the other models across multiple evaluation metrics. When evaluated on DAVIS (Pont-Tuset et al., 2017), SwinTexCo achieves state-of-the-art performance in all metrics except for CDC. Although FAVC (Lei & Chen, 2019) achieves the best result on the CDC metric, its performance in producing vibrant and vivid colors

is poor compared to other methods, leading to a lack of significant meaning of video colorization. Besides, SwinTexCo demonstrates competitive results on Videvo (Lai et al., 2018), surpassing other methods in terms of PSNR and FID while showing slightly inferior performance in SSIM, LPIPS, and CDC. Although not entirely dominant across all metrics on Videvo, SwinTexCo consistently maintains strong video colorization capabilities in long videos.

Fig. 7 shows the outcomes produced by other models compared to SwinTexCo at the 1st, 20th, and 50th frame in the video “horsejumpstick” extracted from the DAVIS evaluation dataset. It is evident from these figures that SwinTexCo excels in generating vivid and vibrant results while maintaining color consistency throughout the videos.

5.2.2. Qualitative results

We select ten videos randomly from the DAVIS dataset and survey 500 individuals. Each participant is required to rate the quality of the colorized output produced by each model on a visual appearance scale ranging from 1 (bad colorized output) to 5 (good colorized output). To ensure a fair result between the approaches, we choose the ground truth of the first frame in each video as a reference image for three models. As mentioned in Section 1.1.3, TCVC (Zhang et al., 2023) consisted of 2 stages: the first is for self-colorizing the first frame in the video, and the latter is for transferring the colors from the result in “stage 1” to the remained frames in the videos. Because of the reason above, we removed “stage 1” and only used “stage 2” in TCVC.

Fig. 8 shows the survey results regarding users’ opinions on the colorized videos generated by three models. This result indicates that SwinTexCo outperforms DeepExemplar (Zhang et al., 2019) and TCVC (Zhang et al., 2023) regarding vibrant and realistic colorization on users’ opinions.

5.2.3. Exemplar-based colorization ability

We also generate images using multiple reference images to compare the exemplar-based colorization ability of SwinTexCo with DeepExemplar (Zhang et al., 2019) and TCVC (Zhang et al., 2023). Fig. 9 shows the outcomes of 3 models with their corresponding reference images at each bottom-left corner. TCVC fails to accurately transfer colors from a reference image to a grayscale image due to slight color inconsistencies in certain areas, resulting in incomplete colorization. The reason for this comes from TCVC’s model training strategy. The model is trained using colorized frames from videos as reference images, which leads to poor coloring performance when using a completely different reference image. While DeepExemplar generally has more vibrant color



Fig. 7. Comparison of colorization between other models and ours at the 1st, 20th, and 50th frames in the video “horsejump-stick” of the DAVIS dataset.

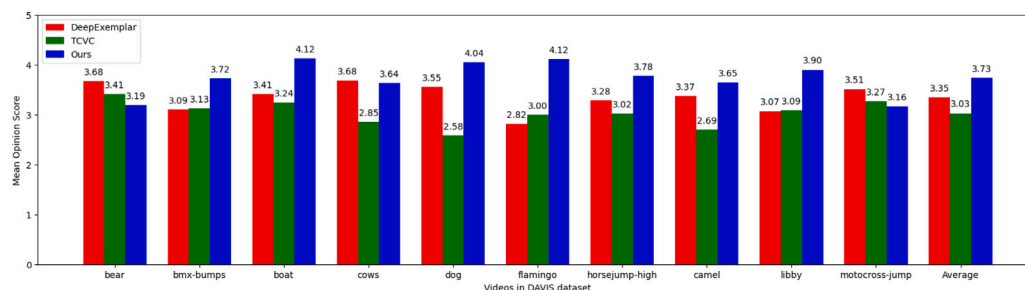


Fig. 8. Evaluation results obtained from users' opinions.

results, the model struggles to differentiate colors between objects, often bleeding color from one object into another, which compromises the image's overall harmony. This is likely due to DeepExemplar's

CNN architecture, which excels at capturing local features but fails to understand the global context in the image. SwinTEExCo outperforms TCVC regarding color transferring ability and DeepExemplar regarding

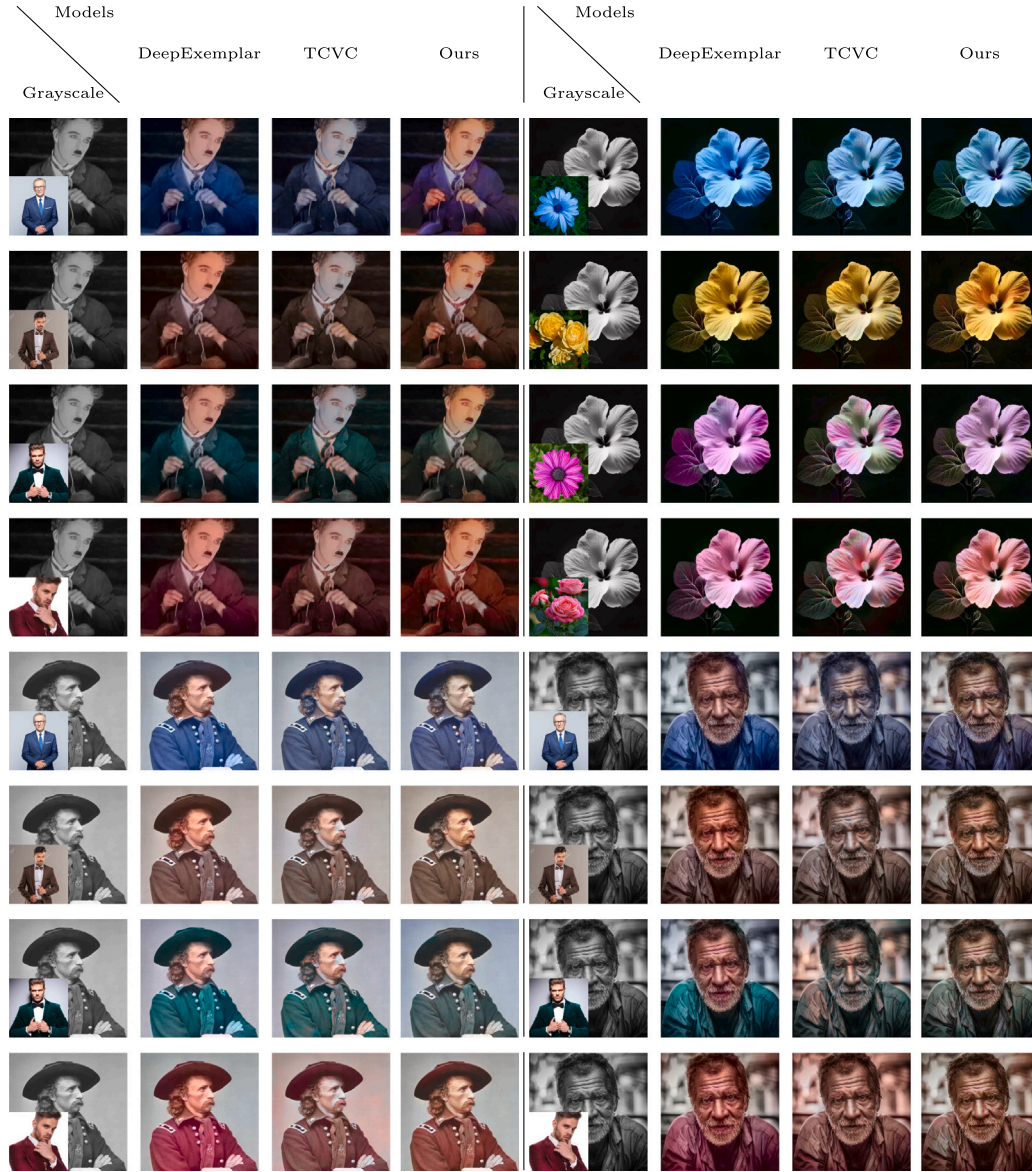


Fig. 9. Different reference images for colorizing the same image. Corresponding reference images are shown at the bottom-left corner of each sample.

the ability to distinguish colors between objects in an image. It demonstrates that the model possesses a remarkable aptitude for adapting colors in a highly flexible manner and accurately colors areas and objects between the input and reference images due to its ability to balance local and global information.

5.2.4. Computational complexity

We analyze and compare the computational complexity of SwinTExCo with other state-of-the-art colorization models: FAVC (Lei & Chen, 2019), Color2Embed (Zhao et al., 2021), VCGAN (Zhao, Po, Yu et al., 2023), DeepRemaster (Iizuka & Simo-Serra, 2019), DeepExemplar (Zhang et al., 2019), TCVC (Zhang et al., 2023) and Vitexco (Tran et al., 2023). The analysis results are collected by inferring the models on the same videos and the same reference image using the NVIDIA Tesla T4 in the same environment. Table 4 shows the number of parameters, the number of floating-point operations (FLOPs) required for single-frame calculation, and the throughput of each model (Frame Per Second or FPS). The results show that SwinTExCo costs the lowest FLOPs with 84.99 GFLOPs and achieves an impressive throughput of 23.7 frames/second. When compared with two exemplar-based models,

SwinTExCo requires 2.03 times fewer calculations than DeepExemplar and 7.07 times fewer than TCVC, while achieving a throughput that is 9.88 times higher than DeepExemplar and 10.30 times higher than TCVC. Furthermore, compared to the prior model Vitexco, SwinTExCo surpasses Vitexco in terms of FLOPs and FPS despite Vitexco having fewer parameters. This advantage arises from the superior Swin Transformer backbone compared to Vitexco's Vision Transformer. Specifically, Swin Transformer utilizes shifted windows to decrease the computational cost of self-attention, resulting in lower FLOPs than Vision Transformer. Additionally, the hierarchical structure and efficient attention mechanism of the Swin Transformer contribute to faster inference and higher FPS. These findings suggest that SwinTExCo offers a competitive advantage in terms of computational efficiency. It can be considered more efficient and scalable by achieving comparable performance with a lower computational cost.

5.3. Ablation study

5.3.1. Analysis of submodules

In order to demonstrate the importance of the submodules utilized in our proposed model, we conduct a comparison by excluding each



Fig. 10. Examples of the ablation of each submodule in the colorization process.

Table 4

Comparison of the computational complexity for single frame colorization.

Model	#params	FLOPs ↓	FPS ↑
FAVC (Lei & Chen, 2019)	24.03M	141.98G	6.4
Color2Embed (Zhao et al., 2021)	71.96M	133.90G	12.4
VCGAN (Zhao, Po, Yu et al., 2023)	66.79M	89.71G	5.4
DeepRemaster (Iizuka & Simo-Serra, 2019)	64.16M	1,611.47G	1.1
DeepExemplar (Zhang et al., 2019)	59.74M	172.12G	2.4
TCVC (Zhang et al., 2023)	124.32M	600.87G	2.3
Vitexco (Tran et al., 2023)	42.85M	322.11G	4.8
SwinTExCo (ours)	78.66M	84.99G	23.7

Table 5

Ablation study for submodules on DAVIS (Pont-Tuset et al., 2017) and Videvo (Lai et al., 2018) datasets.

Experiments	DAVIS dataset				
	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	CDC ↓
w/o Warp	26.96056	0.91821	0.13454	102.13361	0.00457
w/o Colorization	25.64507	0.88071	0.16643	94.82468	0.00381
Full model	32.24419	0.96586	0.06328	56.66980	0.00398
Experiments	Videvo dataset				
	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	CDC ↓
w/o Warp	27.75451	0.93454	0.14090	99.75792	0.00212
w/o Colorization	25.46445	0.88760	0.19682	92.43802	0.00169
Full model	31.96999	0.94401	0.10871	43.36435	0.00172

submodule (Warp or Colorization) separately and analyzing the model results. When the Warp submodule is not used, we implement a nearest-neighbor matching approach, as detailed in DeepExemplar (Zhang et al., 2019). As for the Colorization submodule, we directly map the extracted Correlation Matrix $\mathcal{M}$ and Similarity Map $\mathcal{S}$ with the input grayscale frame to generate the color frame, without passing them through the Colorization Submodule. The SwinT Extractor submodule is not analyzed, as its effectiveness has been demonstrated through the comparison between Vitexco (Tran et al., 2023) and SwinTExCo in Table 3.

The results of the submodule analysis are presented in Table 5 and Fig. 10. We observed that the absence of each submodule leads to a noticeable decline in the quality of the colorization process. Without the Warp submodule, there is a significant decrease in statistical results. Although the nearest neighbor matching algorithm is effective, it is inadequate for determining the similarity between the extracted features from the video frames and the reference image. Similarly, the quality of the colorization process suffered when the Colorization submodule is not utilized, with most statistics showing a decline, except for the CDC metric. This can be explained by the fact that the CDC metric measures the consistency of color distribution in colorized frames and thus performs well when adjacent frames in a video have limited color. However, other metrics, especially PSNR, indicate poor results, suggesting that the frames were not effectively colorized.

5.3.2. Analysis of loss functions

We evaluated the importance of each loss function by individually removing them and assessing their impact on the colorization process. We conducted the ablation study on four loss functions: L_{ctx} , L_{perc} ,

Table 6

Ablation study for loss functions on DAVIS (Pont-Tuset et al., 2017) and Videvo (Lai et al., 2018) datasets.

Experiments	DAVIS dataset				
	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	CDC ↓
w/o L_{ctx}	26.92999	0.91706	0.13157	108.01889	0.00511
w/o L_{perc}	26.66825	0.91486	0.13538	102.01840	0.00459
w/o L_{smooth}	26.65985	0.91071	0.13649	97.18691	0.00418
w/o L_{temp}	26.48000	0.91273	0.13578	103.45673	0.00616
All loss functions	32.24419	0.96586	0.06328	56.66980	0.00398
Experiments	Videvo dataset				
	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	CDC ↓
w/o L_{ctx}	27.74750	0.93467	0.13923	101.23100	0.00231
w/o L_{perc}	27.60584	0.93276	0.13951	94.28251	0.00205
w/o L_{smooth}	27.51363	0.93017	0.14285	88.25016	0.00196
w/o L_{temp}	27.54234	0.93085	0.14092	91.75870	0.00288
All loss functions	31.96999	0.94401	0.10871	43.36435	0.00172

L_{smooth} , and L_{temp} . The remaining two loss functions were not analyzed due to technical reasons. Without the L_{L1} loss, the model cannot learn how to transfer color from the reference image to the grayscale video frames. The L_{adv} loss is calculated from the output of the Discriminator module to evaluate the colorization process in the mechanism of a Generative Adversarial Network (GAN). If we remove this loss function, the overall model cannot learn the colorization process. We do this by training the model using the weighted loss function mentioned in Formula (8), removing each of the above loss functions, and keeping all other remaining loss functions. The results of the loss functions analysis are presented in Table 6 and Fig. 11.

In general, the performance of the proposed model declines when any of the above loss functions are removed. When we remove L_{ctx} , the performance of the colorization process, as evaluated by FID metrics, is significantly affected. This is because L_{ctx} focuses on contextual similarity, and FID is a metric for evaluating the similarity between the features of the ground truth frame and the colorized frame, which contains both information on colors and context. Similarly, when we remove L_{temp} , the performance of the colorization process, as evaluated by CDC metrics, is significantly affected. This is because L_{temp} focuses on color changes in the video frames, and CDC is a metric for measuring color consistency through video frames.

6. Conclusion

This research introduces SwinTExCo, the first Swin Transformer exemplar-based video colorization algorithm featuring a recurrent architecture. Our approach not only achieves temporal consistency but also generates realistic predictions in video colorization. Both quantitative and qualitative metrics demonstrate that SwinTExCo outperforms existing approaches. Notably, our method enables video colorization based on a reference image. The accuracy of the video colorization increases when utilizing a larger reference image dataset. In our future work, we aim to improve the model's quality by expanding the size and input resolution. Additionally, we plan to explore integrating bidirectional recurrent methods to improve temporal consistency. The source code and data can be accessed at: <https://github.com/Chronopt-Research/SwinTExCo/>



Fig. 11. Examples of the ablation of each loss function in the training process.

CRediT authorship contribution statement

Duong Thanh Tran: Conceptualization, Methodology, Software, Validation, Data curation, Writing – original draft. **Nguyen Doan Hieu Nguyen:** Conceptualization, Methodology, Software, Validation, Data curation, Writing – original draft. **Trung Thanh Pham:** Conceptualization, Methodology, Software, Validation, Data curation, Writing – original draft. **Phuong-Nam Tran:** Conceptualization, Methodology, Software, Writing – original draft. **Thuy-Duong Thi Vu:** Conceptualization, Methodology, Supervision. **Cuong Tuan Nguyen:** Conceptualization, Methodology, Writing – original draft, Supervision. **Hanh Dang-Ngoc:** Writing – review & editing. **Duc Ngoc Minh Dang:** Conceptualization, Methodology, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared data and source code in the revised manuscript.

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